

The $\mathbb{R}^3$ cone has the metric

$$ds^2 = dr^2 + r^2(dx^2 + dy^2),$$



r - is the radial coord along the cone

Σ : $(n-1)$ dim embedded, closed C^∞ mfd.

Def: The topological cone $C(\Sigma)$ over Σ is defined as

$$C(\Sigma) := ([0, \infty) \times \Sigma) / (\{0\} \times \Sigma)$$

$\rightarrow$ What is this Quotient?

$$C(\Sigma) = \{ [t, x] \in [0, \infty) \times \Sigma \}$$

Vertex $\hat{=} 0$

For $r > 0$, $C_r(\Sigma) = \{ [t, x] \in C(\Sigma) \mid t < r \}$.

why is $t < r$? well in 2D:



etc...

Def: Let M be a C^∞ mfd w/ isolated singularities

S. A Riemannian metric g on $M \setminus S$

is called C^0 -conical, resp. C^2 -conical,

at a singular pt $p \in S$, if $\exists$ a Conical Chart $(C_p(\Sigma_p), \varphi_p)$ and a Riemannian metric h_p on Σ_p s.t.

i) $\varphi_p^* g = dt^2 \oplus h_t$

ii) $\frac{1}{t} h_t \rightarrow h_p$, $\frac{1}{2t} \frac{\partial}{\partial t} h_t \rightarrow h_p$

or

iii) $\varphi_p^* g = dt^2 \oplus h_t$, $\frac{1}{t} h_t \rightarrow h_p$, $\frac{1}{2t} \frac{\partial}{\partial t} h_t \rightarrow h_p$

$$\frac{1}{2} \frac{\partial^2}{\partial t^2} h_t \rightarrow h_p \text{ on } C^2 - \text{top.}$$

res.

$$\varphi_p: (C_p(\Sigma_p), \sigma) \xrightarrow{\cup \Sigma_p} (U_p, \rho)$$

$$\varphi_p(\sigma) \equiv p$$

Def: Let (Σ, g) be a Riem. mfd of even n.
 (Σ, g) is Sasakian if the holonomy group
of the metric cone on Σ , $C(\Sigma)$
 $(C(\Sigma), \bar{g}) = (\mathbb{R}^+ \times \Sigma, dr^2 + r^2 g)$
reduces to a subgroup $U(\frac{n}{2})$
 $m = 2n+1, n \geq 1 \Rightarrow (C(\Sigma), \bar{g})$ is Kähler.

Prop: Let (Σ, g) be a Riem. Mfd. The
Levi-Civita conn ∇ ; and let $R(x, y)$
 $R(x, y): T(\Sigma) \rightarrow T(\Sigma)$. Then
(i) $\exists$ a Killing vec. h of unit length
on Σ s.t. the tensor field Φ
of type $(1,1)$ defined by

$$\Phi(x) := \nabla_x h$$

Satisfies:

$$(\nabla_x \Phi)(y) = g(h, y) x - g(x, y) h$$

$$\forall x, y \in T\Sigma$$

(ii) (Σ, g) is Sasakian.

* There is a natural splitting of the tangent bundle

$$TC(\Sigma) = \mathcal{L}_\psi \oplus \mathcal{L}_\mu \oplus \mathcal{H}$$

$\mathcal{L}_\chi :=$ trivial line bundle generated by the nowhere vanishing vec. field χ

• Let I be a complex structure on $C(\Sigma)$,

$$\nabla I = 0 \Rightarrow \psi = r \partial_r$$

ψ : Euler field on $C(\Sigma)$

$$I\psi := \mathbb{E}(\psi) - n(\psi)\psi = I\psi = \psi$$